Soumyadyuti Ghosh

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Education & Research

■ **New York University Abu Dhabi (NYUAD), Abu Dhabi, UAE** **2025 – Present**
Postdoctoral Associate, Center for Cyber Security (CCS)
Supervisor: Prof. Michail Maniatakos

Primary area of work: i) Privacy-preserving computation using Fully Homomorphic Encryption (FHE), with emphasis on encrypted keyword search and abuse detection in end-to-end encrypted messaging, and high-accuracy encrypted arithmetic for smart grids; ii) Class prior-free Positive-unlabeled learning, and practical validation of learning models on real-world fine-grained datasets.

■ **Indian Institute of Technology Kharagpur, India** **2019 – 2025**
Ph.D. in Computer Science and Engineering
Supervisors: Prof. Debdeep Mukhopadhyay and Prof. Soumyajit Dey
Secured Embedded Architecture Laboratory (SEAL) and High Performance Real-Time Computing Laboratory (HiPRC)

Area of work: Privacy-preserving computation in Cyber-Physical Systems (CPS), primarily in the Smart Grid domain. Research focuses on identifying vulnerabilities and developing methodologies for secure communication and privacy-preserving computations, while ensuring system stability through cryptographic and information-theoretic techniques such as Differential Privacy, Secure Multi-Party Computation, and Fully Homomorphic Encryption.

■ **Jadavpur University, West Bengal, India** **2012 – 2016**
B.E. in Computer Science and Engineering
Graduated with 7.83 CGPA

- Secured **364$^{\text{th}}$ rank** out of **150,000+** students in **Joint Entrance Examination**, 2012.
- **Higher Secondary:** 86.2% overall and 94.34% in science category, W.B.C.H.S.E, India, 2012.
- **Secondary:** 82% overall and 96.34% in science category, W.B.B.S.E, India, 2010.

Work Experience

■ **Wipro Digital, Bangalore, India** **2016 – 2019**
Senior Developer and Project Lead

- Designed and developed a PaaS model with a secure end-to-end communication gateway for medical devices, supporting device registration, data synchronization, remote access, file transfer, network bandwidth throttling, concurrency management, priority handling, and logging.
- Developed an automated robot capable of identifying random objects in industrial environments and performing essential operations using the Microsoft Azure platform.
- Integrated soil moisture sensors to determine critical moisture levels and maintained real-time time-series data on Bluemix and GE Predix services.

Research & Internship Experience

■ **Visiting Research Scholar, Center for Cyber Security (CCS), New York University Abu Dhabi (NYUAD), 2023**
Supervisor: Prof. Michail Maniatakos

- Researched accurate floating-point division in the CKKS homomorphic encryption scheme, applied to computations on encrypted Smart Grid data, enabling secure and precise grid functionalities while preserving confidentiality.

■ **Summer Internship, Department of Computer Science and Engineering, Indian Institute of Technology (IIT) Kharagpur, 2015**
Supervisor: Prof. Chittaranjan Mandal

■ **Project Internship, Department of Computer Science and Engineering, Jadavpur University, Kolkata, 2016**
Supervisor: Prof. Chandan Mazumdar

Selected Research Projects & Tools

■ **Cyber Security Research in Cyber-Physical Systems**
Sponsored by Haldia Petrochemicals Ltd. and TCG Foundation

- **Smart-Grid Security:** Set up a Hardware-in-the-Loop testbed with OPAL-RT simulator, CompactRIO controller, PMUs, PDC, V/I Power Amplifier, GPS Clock, and automation controller to study the impact of various attacks on power system control loops and propose suitable countermeasures, including secure communication protocols in the smart grid environment.

■ **Next-Generation Secured Internet of Things (IoT)**
Funded by the Department of Science and Technology (DST), Government of India

- **Smart Meter Design:** Developed a low-cost, secure smart meter on Raspberry Pi 4 for integration into a smart grid testbed. Embedded a spoofing-resistant PUF-based authenticated key exchange (AKE) protocol. Simulated environmental changes using a CME bench-top chamber to assess SRAM-PUF reliability and evaluate AKE protocol performance.

■ **Centre on Hardware Security**
Data Security Council of India and Ministry of Electronics and Information Technology, Government of India

- **Assessment of PLCs used in Smart Grids:** Used CompactRIO as PLC and OPAL-RT as plant simulator to realize closed-loop grid functionalities and study vulnerabilities of PLCs under various attack scenarios (e.g., HARVEY, NDSS'17), providing insights into operational challenges in realistic Smart Grid environments.

■ **KAVACH: Key Analysis and Vulnerability Assessment for Cryptographic Hardware**
Co-developed with PROC SHIELD Private Limited

- Co-developed a GUI-based platform for side-channel and fault analysis of cryptographic hardware. KAVACH controls FPGAs and oscilloscopes, captures power and electromagnetic traces, and integrates ChipWhisperer Lite for trace acquisition and fault injection. It supports Correlation Power Analysis for AES-128 key recovery and clock-glitch experiments for Differential Fault Analysis. By integrating acquisition, analysis, and fault-injection controls in one interface, the tool supports repeatable hardware-security evaluation.

Publications

■ Accepted Publications

- "Demand Manipulation Attack Resilient Privacy Aware Smart Grid Using PUFs and Blockchain." In **Applied Cryptography and Network Security Workshops (ACNS)**, 2021.
- "Is the whole lesser than its parts? Breaking an aggregation based privacy aware metering algorithm." In **25th Euromicro Conference on Digital System Design (DSD)**, 2022.
- "SMarT: A SMT Based Privacy-Preserving Smart Meter Streaming Methodology." In **International Conference on Security, Privacy, and Applied Cryptography Engineering (SPACE)**, 2022.
- "hello? is there anybody in there? Leakage assessment of differential privacy mechanisms in smart metering infrastructure." In **International Conference on Applied Cryptography and Network Security (ACNS)**, Springer, 2024.
- "Differentially Private Real Time Pricing Control for Smart Grids." **ACM Transactions on Cyber-Physical Systems (TCPS)**, 2025.
- "Pay What You Spend! Privacy-Aware Real-Time Pricing with High Precision IEEE 754 Floating Point Division." In **ACM Asia Conference on Computer and Communications Security (ASIACCS)**, 2025. [Selected for contributed talk in TPMPC]
- "HARP: Privacy-Preserving High-Accuracy Real-Time Pricing with Precise Encrypted Inverse for Smart Grids." In **European Symposium on Research in Computer Security (ESORICS)**, 2026.
- "A Learning-Enabled Framework for Profit-Driven Attack Synthesis in Electricity Markets." In **International Conference on Security, Privacy, and Applied Cryptography Engineering (SPACE)**, 2026.
- "PULSE: Support-Calibrated Class Prior-Free Positive-Unlabeled Learning with Hard-Negative Ranking and Lower-Tail Coverage." In **ACM International Conference on Information and Knowledge Management (CIKM)**, 2026.

■ Under Review Publications

- "Privacy-Aware and Stable Control in Smart Grids with Intentionally Skipped Executions."
- "Elasticity-Aware Predictive Ensemble for Customer-side Non-Technical Loss Detection."
- "Secure and Efficient Encrypted Keyword Search For Privacy-Preserving Messaging Protocols."
- "Private Similarity-Threshold Abuse Detection in Encrypted Messaging."

Professional & Academic Service

- **Talks, Tutorials, and Seminars:**
  - Talk at the **Workshop on Architecture for Security and Privacy (WASP)**, Indian Institute of Technology Kharagpur, 2024.
  - Tutorial lecture on Secure Multi-Party Computation (MPC) and Fully Homomorphic Encryption (FHE) in the **Privacy Enhancing Technologies (PETs)** course at Indian Institute of Technology Kharagpur, 2025.
  - Invited guest lecture at Boot Camp, **Indian Institute of Technology Kanpur**, 2025.
  - Invited contributed talk at the Workshop on **Theory and Practice of Multi-Party Computation (TPMPC)**, 2025.
  - Seminar talk as part of the **CCSAD Cybersecurity Seminar Series, New York University Abu Dhabi (NYUAD)**, 2025.
- **Teaching Assistant (Indian Institute of Technology Kharagpur):** Cryptography and Network Security, Privacy Enhancing Technologies (PETs), Computational Foundations of Cyber-Physical Systems, High-Performance Computer Architecture, Programming and Data Structures (theory and lab), Computer Organization Lab, Formal Language and Automata Theory, Hardware Security.
- **Journal and Proceedings Reviewer:** IEEE TCAD, IEEE TSG, IEEE TIFS, IEEE TDSC, IEEE PES Letters, ACM TECS, ACM TCPS.
- **Conference and Workshop Reviewer:** USENIX Security, NDSS, CHES, ICCAD, DATE, DAC, ACNS, DSD, HOST, SPACE, ASHES, AsianHOST.
- **Event Organization and Service:**
  - Contributed to the organization of the **Workshop on Cyber-Physical System Security**, Indian Institute of Technology Kharagpur, India, 2019.
  - Contributed to the organization of the International Conference on **Security, Privacy, and Applied Cryptography Engineering (SPACE)**, Indian Institute of Technology Kharagpur, India, 2021.
  - Contributed to **workshop on Architecture for Security and Privacy (WASP)**, Indian Institute of Technology Kharagpur, 2024.
  - Contributed to **workshop on Machine Learning and Hardware Security**, Indian Institute of Technology Kharagpur, 2025.
  - Supported the Middle East and North Africa (MENA) regional activities of **Cybersecurity Awareness Worldwide (CSAW)**, organized by the Center for Cyber Security at NYUAD, including serving as a judge and evaluator for the BioHack 3D competition.

Skills

Programming: C, C++, Java, Python, MEAN and MERN stack, MapReduce.

Technical:

- **Differential Privacy:** Google's differential privacy libraries.
- **Secure Multi-Party Computation:** SPDZ, MP-SPDZ, SCALE-MAMBA, SecFloat, MOTION, Blaze, Swift, EzPC, Falcon, Bristol Fashion MPC Circuits, CBMC-GC.
- **Homomorphic Encryption:** Microsoft SEAL, node-seal, OpenFHE, HEAAN, and Pyfhel.
- **Machine Learning:** Machine learning and deep learning model development, weakly supervised learning, predictive and ensemble modeling, performance evaluation, and validation on real-world datasets using Python, PyTorch, NumPy, SciPy, and scikit-learn.
- **Foundations of Cyber-Physical Systems (CPS):**
  - Stochastic modeling and simulation of CPS.
  - Closed-loop stability analysis of CPS dynamics using discrete-time models, characteristic equations, and dominant-pole analysis in the z-plane.
  - MATLAB, Simulink, GridLAB-D, YALMIP, and multidimensional Kalman filtering.
- **Side-channel and fault analysis** of cryptographic hardware, including power and electromagnetic trace acquisition, Correlation Power Analysis, Differential Fault Analysis, and clock-glitch fault injection. Hardware-security experimentation using FPGAs, oscilloscopes, ChipWhisperer Lite.
- **Satisfiability Modulo Theories:** Z3 solver.
- **Floating-Point Arithmetic:** IEEE-754, POSIT.

Academic Courses: Cryptography and Network Security, Computational Foundations of Cyber-Physical Systems, Data Structure and Algorithms, Computer Architecture, Digital Logic, Operating Systems, Computer Networks, Formal Methods, Automata Theory, Database Management, Advanced Machine learning.

Cloud Computing: Microsoft Azure, Casablanca (Azure REST SDK), Amazon Web Services, IBM Bluemix, GE Predix.

Hardware Expertise: OPAL-RT real-time simulator; National Instruments CompactRIO Real-Time Controller (CRIO 9076); Phasor Measurement Units (SEL 421, SEL 2240); Phasor Data Concentrator (SEL-3555); V/I Power Amplifier; GPS Clock (SEL 2404); SEL SynchroWAVE Historian; Power Meters (PM 5320, PM 5330); Raspberry Pi 4 Model B; ChipWhisperer Lite.

Awards, Certifications & Extracurricular Activities

- Second place in the 'Embedded Security Challenge" at Cyber Security Awareness Week (CSAW) in 2020 and 2021.
- Topper in the Ph.D. comprehensive examination at IIT Kharagpur.
- Recipient of the Wipro Digital 'Wunderkind" Award, 2018.
- IBM Certified Application Developer – Cloud Platform V1.
- Secured fourth place at the state level in the Mathematics Olympiad, 2008.
- Regular chess player on Chess.com with a rapid rating of 2267.
- Competitive Auction and Duplicate Bridge player, runner-up in the college Bridge tournament in 2015 and winner in 2016.

References

- **Prof. Debdeep Mukhopadhyay**
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- **Prof. Soumyajit Dey**
  Department of Computer Science and Engineering,
  Indian Institute of Technology Kharagpur, India
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- **Dr. Ajith Suresh**
  Technology Innovation Institute (TII), Abu Dhabi, UAE
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I hereby declare that the above particulars are true and correct to the best of my knowledge.

Date: August 31, 2026 **Soumyadyuti Ghosh**